

Equipment Performance Evaluation/AC & HP Start Up Report

Job Name: _____ Date: _____
Address: _____ Phone: _____
City: _____ State: _____ Zip: _____

Company Name _____ Company Phone: _____
Technician: _____ Technician Cell: _____
Dealer # _____

Complaint/Symptoms/Visual Inspection (Details, don't leave anything out, pictures can also be very beneficial when reporting an issue, use last page for additional space if needed)

Equipment Information

	Outdoor Unit	Indoor Coil	Furnace (or Blower Coil)	Accessory	Thermostat
Manufacturer					
Model #					
Serial #					
Date Installed					
Lineset Length: _____ Rise: _____ Suction Line Size: _____ Liquid Line Size: _____ Number of Elbows: _____					
Indoor Coil is ☐ Above ☐ Below ☐ Same Level as Outdoor Unit Matched System yes no (circle one)					

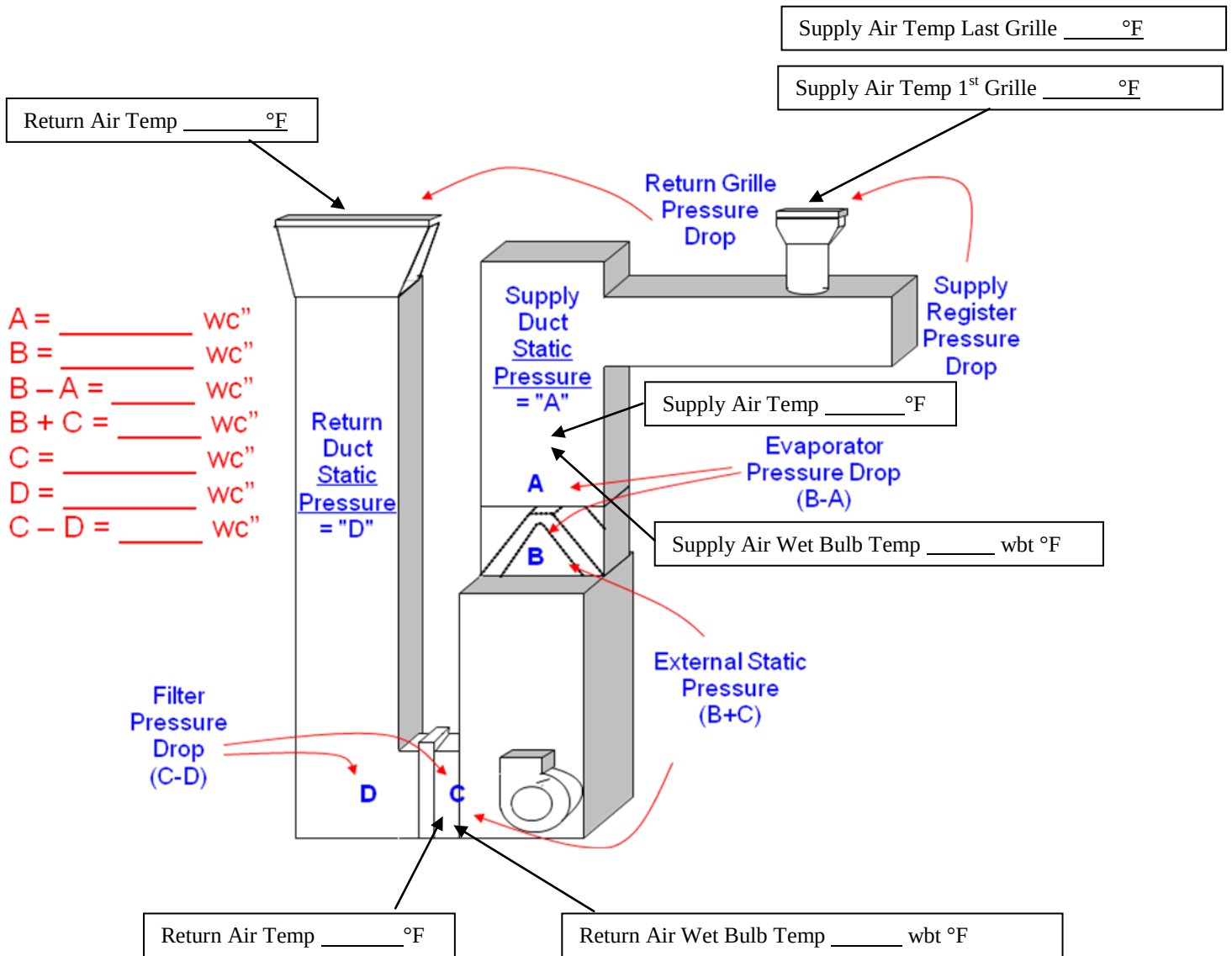
Cooling/Heating Performance AC or HP & Air Handler (Complete the sections that apply to your issue)

(circle Cool or Heat)	Cool / Heat	Cool / Heat	Cool / Heat
Suction psig @ ODU	/	L/Line psig @ OD unit	/
Line Temp @ Vapor S Valve	/	L/L Temp @ OD Unit °F	/
Vapor Line @ IDU °F	/	Subcooling °F	/
Superheat @ IDU °F	/	Required Subcool / Approach	/
Superheat @ ODU °F	/	L/L Temp @ ID Unit °F	/
Return Air Dry Bulb °F	/	Discharge Line Pressure	/
Supply Air Dry Bulb °F	/	Discharge Line °F	/
Dry Bulb Delta T °F	/	OD Ambient Dry Bulb °F	/
Supply Air Wet Bulb °F	/ NA	Return Static "WC	/
Return Air Wet Bulb °F	/ NA	Supply Static "WC	/
Indoor Metering Device ☐ TXV (# 's _____) ☐ Fixed Orifice (Size _____)			

Heating Performance – Gas Furnace (Complete the sections that apply to your issue)

Vent: ☐ 2 Pipe ☐ 1 Pipe ☐ B-Vent Length _____ # Elbows _____	Vent Termination: ☐ Concentric or ☐ Two-Pipe or ☐ One-Pipe		
Fuel Type (Nat / LP) circle one	Percent ☐ CO ₂ or ☐ O ₂	Flame Signal µA	
Static Line Pressure "WC (Nat / LP)	CO ppm	Temperature Rise °F	
Dynamic Line Pressure "WC (Nat/LP)	Return Static "WC	Nameplate Temp Rise	
Manifold Pressure "WC (Nat / LP)	Supply Static "WC	Trans. Secondary Voltage	
Supply Air DBT °F @ unit/sup grille	Total External Static "WC	ID Line Voltage	
Return Air DBT °F @unit / return grille	OD Temperature °F	ID Blower Amps	
Other:	CFM (see formula Pg. 2)	Other:	

AIRFLOW DIAGNOSTICS WORKSHEET

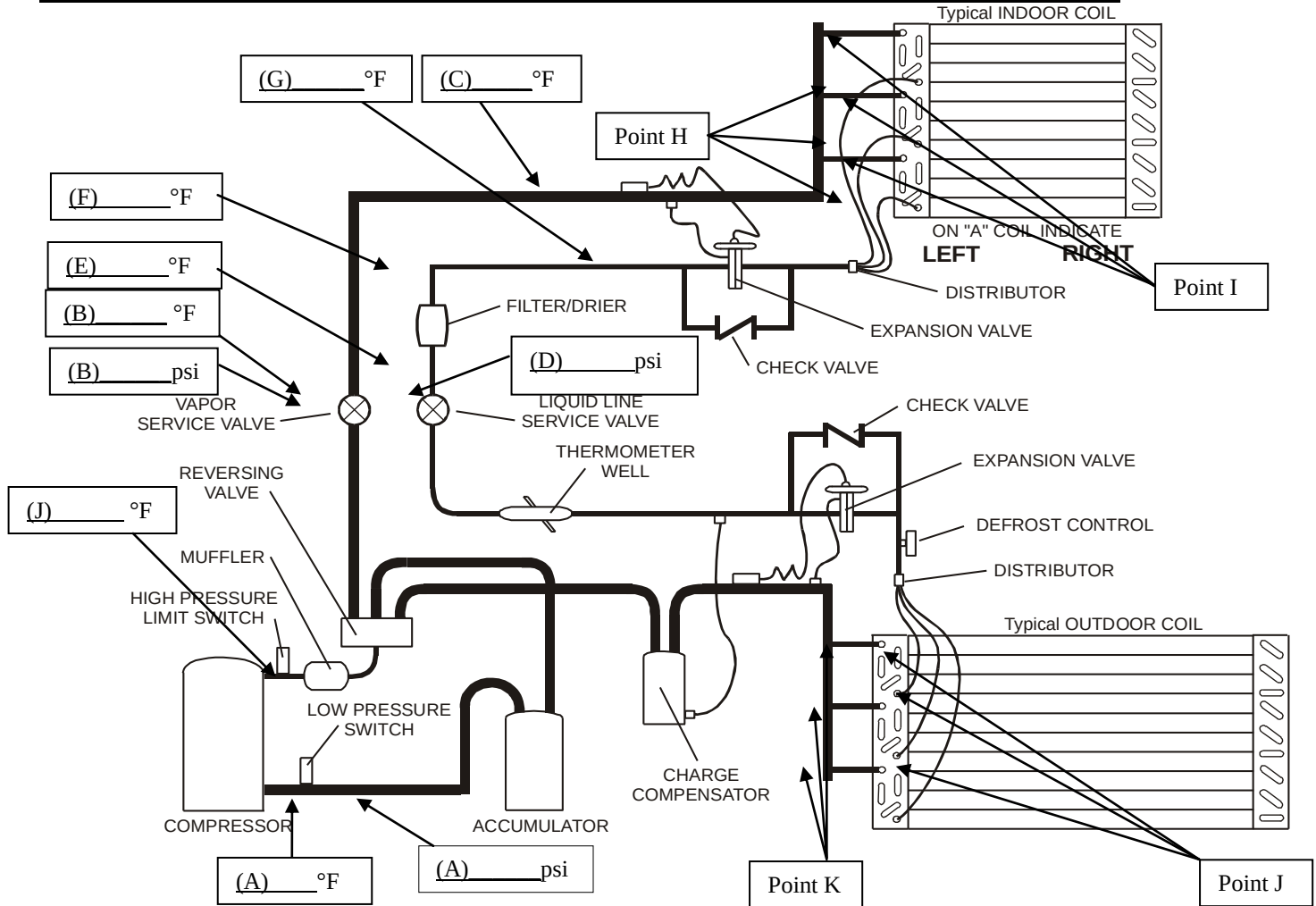


CFM FORMULAS (Sensible Heat)

Air Handlers: $CFM = \frac{\text{Volts} \times \text{Amps} \times 3.413}{\Delta T \times 1.08}$
 Allow the appliance to operate until the temperature rise stabilizes. Measure the temperature rise again out of the line of sight of the electric heater, along with the incoming volts and current draw in amps to the electric strip heaters. Enter the information into the following formula.

Gas Furnace: $CFM = \frac{Btuh}{\Delta T \times 1.08}$
 $Btuh = (\text{Input BTU} \times \text{thermal efficiency}) / (1.08 \times \Delta T)$
 Combustion efficiency can be used in place of thermal efficiency.

REFRIGERATION DIAGNOSTICS WORKSHEET



COOLING MODE

(A) Suction Line Temp _____ °F
 (A) Suction Pressure _____ psi
 Superheat @ Compressor _____ °F
 (B) Suction Line Temp _____ °F
 (B) Suction Line Pressure _____ psi
 (C) Suction Line Temp _____ °F
 Superheat @ ID TXV _____ °F
 (D) Liquid Line Press _____ psi
 (E) Liquid Line Temp _____ °F
 (F) Liquid Line Temp _____ °F
 Subcooling @ ODU _____ °F
 Temp Drop Across Drier _____ °F
 (G) Liquid Line Temp IDU _____ °F
 Subcooling @ IDU _____ °F
 (H) Dist Cir 1 Line Temp _____ °F
 (H) Dist Cir 2 Line Temp _____ °F
 (H) Dist Cir 3 Line Temp _____ °F
 (H) Dist Cir 4 Line Temp _____ °F
 (I) Cir 1 Return Line Temp _____ °F
 (I) Cir 2 Return Line Temp _____ °F
 (I) Cir 3 Return Line Temp _____ °F
 (I) Cir 4 Return Line Temp _____ °F
 (J) Discharge Line Temp _____ °F

HEATING MODE

(A) Suction Line Temp _____ °F
 (A) Suction Pressure _____ psi
 Superheat @ OD TXV _____ °F
 (B) Discharge Line Temp _____ °F
 (B) Discharge Pressure _____ psi
 (D) Liquid Line Press _____ psi
 (E) Liquid Line Temp _____ °F
 (F) Liquid Line Temp _____ °F
 Temp Drop Across Drier _____ °F
 (G) Liquid Line Temp IDU _____ °F
 (J) Dist Cir 1 Line Temp _____ °F
 (J) Dist Cir 2 Line Temp _____ °F
 (J) Dist Cir 3 Line Temp _____ °F
 (J) Dist Cir 4 Line Temp _____ °F
 (K) Cir 1 Return Line Temp _____ °F
 (K) Cir 2 Return Line Temp _____ °F
 (K) Cir 3 Return Line Temp _____ °F
 (K) Cir 4 Return Line Temp _____ °F

NOTES:

Please list all jumpers removed and/or clipped. Also show all jumper/dip switch settings on any and all circuit boards on equipment and/or accessories. Plus any additional comments about operation that you observe.

[illegible]